THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** drop objects of different masses from the same height	Test by a controlled experiment.
2	**(a)** pumpkin: creeper	Pumpkin has a weak stem that spreads.
3	**(c)** protein and vitamins	Polished rice mostly has starch.
4	**(d)** either a magnet or magnetic material	Attraction alone does not prove a magnet.
5	**(b)** 120 cm	1.2 x 100 = 120.
6	**(a)** cork and leaf	Cork and leaf are lighter than water.
7	**(c)** fever	Above 37 °C.
8	**(b)** evaporation	Slow vaporisation.
9	**(d)** magnet	Removes iron.
10	**(c)** it sprouts when given water	Germination indicates life.
11	**(d)** forests	Forests can regrow but not if cut faster than they grow.
12	**(b)** Mercury, Venus, Earth, Mars	Order.
13	**(b)** plants need light for healthy growth	Without light, plants turn pale.
14	**(d)** crocodile	Crocodiles live both in water and land.
15	**(d)** vegetables and fruits	They are protective foods.
16	**(c)** iron, nickel, cobalt	Standard magnetic materials.
17	**(b)** a mass vibrating on a spring	It moves to and fro on a straight path.
18	**(d)** the boat's shape holds a lot of air	Large volume with air makes the boat's overall density small.
19	**(d)** because of the kink	Kink holds mercury.
20	**(a)** evaporation and condensation	Water vapour cools.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Then the effect can be linked to that one cause (1); It makes results clear and fair (1).	1+1
22	To reduce loss of water (1); The green stem makes the food (1).	1+1
23	Proteins and vitamins (1); Minerals and fibre are also low (1).	1+1
24	No, a magnet repels one pole (1); The rod is a magnetic material only (1).	1+1
25	3.75 x 1000 = 3750 m (1); 3750 x 100 = 375000 cm (1).	1+1
26	The boat has a large hollow shape with air (1); Water supports it as its average density is low (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Take two sets of identical seeds with same moisture (1); Keep one set in a warm and another in a cool place (1); Compare sprouting after fixed days (1).	1+1+1
28	Lotus: water (1); Cactus: desert (1); Pine: mountain (1).	1+1+1
29	It saves time but lacks nutrients (1); It has too much fat, sugar and salt (1); Regular use harms health (1).	1+1+1
30	Use a magnet to remove iron (1); Add water to dissolve salt, filter sand (1); Evaporate water to get salt (1).	1+1+1
31	240 cm / 15 = 16 (1); 16 pieces exactly (1); Leftover 0 cm (1).	1+1+1
	OR: Magnet removes iron (1); Dissolve sugar in water (1); Filter sand and evaporate water for sugar (1).	1+1+1
32	Heat flows to room (1); Water cools (1); Becomes about 25 °C (1).	1+1+1
	OR: Evaporation from the sea (1); Condensation into clouds (1); Precipitation and flow back (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Change: length of the string (1); Keep same: bob, angle of release (1); Measure time for 10 swings (1); Repeat and average (1); Record in a table and conclude (1).	1+1+1+1+1
34	Camel: hump and broad feet (1); Fish: gills and fins (1); Bird: wings and light bones (1); Polar bear: thick fur and fat (1); Each helps survival in its habitat (1).	1+1+1+1+1
35	Reduce junk food (1); Add fruits and vegetables (1); Add dal or eggs for protein (1); Add milk for calcium (1); Encourage water and exercise (1).	1+1+1+1+1
36	Hang one bar magnet (1); Bring N of other near N (1); Observe repulsion (1); Bring S near N: attraction (1); Conclude: like repel, unlike attract (1).	1+1+1+1+1
	OR: Wrap thread around the glass (1); Measure its length (1); Measure the diameter with a scale (1); Divide circumference by diameter (1); Ratio is nearly 3.14 (1).	1+1+1+1+1
37	Heat water and cover (1); Measure temperature drop after 30 min (1); Same conditions for all (1); The least drop is best insulator (1); Cork or plastic likely (1).	1+1+1+1+1
	OR: Rise = 8 °C each (1); After 2 min: 38 (1); After 4 min: 46 (1); After 10 min = 30 + 5 x 8 (1); 70 °C (1).	1+1+1+1+1
38	Take plates of different size with same water (1); Keep one in sun and one in shade (1); Use a fan for one (1); Note time to dry (1); Large, warm, windy dries fastest (1).	1+1+1+1+1
	OR: Magnet removes iron (1); Add water (1); Sawdust floats and is skimmed (1); Filter sand (1); Evaporate to get sugar (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERKEY

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** unsaturated	More solute can dissolve.
2	**(c)** pupa	Metamorphosis.
3	**(c)** glass bottles	Bottles can be used again.
4	**(c)** complete one revolution around the Sun	Orbital period.
5	**(a)** The number alone has no meaning without it	Units give meaning.
6	**(d)** herb, shrub, tree	Herbs are smallest.
7	**(c)** the list of ingredients	Ingredients tell what we eat.
8	**(c)** push away from each other	Like poles repel.
9	**(a)** hold the tape against the edge of the object so that measuring starts from zero	Hook sets zero.
10	**(d)** food does not stick to the surface	Non-stick.
11	**(a)** 30 °C	Room temperature.
12	**(d)** drip irrigation	Saves water.
13	**(c)** to avoid spurting	Splashing.
14	**(c)** reproduction	Life continues.
15	**(b)** 21 L	About 21 per cent.
16	**(a)** it rotates once in the time it orbits the Earth	Synchronised.
17	**(c)** the presence of water	Only one variable changes at a time.
18	**(a)** herb	Soft stems and low height.
19	**(a)** roti, dal, vegetables and curd	Variety of nutrients.
20	**(c)** north pole	Like poles repel.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Above normal 37 °C (1); The person has a fever (1).	1+1
22	Heat is used to change water into steam (1); It is not used to raise the temperature (1).	1+1
23	Sand does not dissolve (1); Sedimentation and filtration are suitable (1).	1+1
24	It germinates under suitable conditions (1); Its life processes are slow (1).	1+1
25	It contains many organisms (1); It supports plant growth (1).	1+1
26	Day is one rotation (1); Year is one revolution (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Handpick stones (1); Use magnet for nails (1); Sieve out sand (1).	1+1+1
28	Fire does not grow from within (1); It does not reproduce on its own (1); It cannot respond or repair itself (1).	1+1+1
29	Light a candle (1); Cover with a jar (1); It goes out after some time (1).	1+1+1
30	Earth has air and water (1); Moon has none (1); Moon has extreme temperatures (1).	1+1+1
31	Use same size and number of sheets (1); Pour same amount of water on each (1); Compare water absorbed (1).	1+1+1
	OR: Gills (1); Fins and tail (1); Streamlined body (1).	1+1+1
32	Dehydration (1); Wastes not removed (1); Tiredness and headache (1).	1+1+1
	OR: Bring the marked N pole near an end (1); If attracted, that end is S (1); If repelled, that end is N (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Place a potted plant near a window (1); Rotate it after some days (1); It bends towards light (1); Rotate again (1); It bends again (1).	1+1+1+1+1
34	Take equal volumes in funnels (1); Pour equal water (1); Measure water collected (1); Clay holds the most (1); Sand holds the least (1).	1+1+1+1+1
35	Locate Saptarishi (1); Extend line from the bowl (1); Pole star gives north (1); New moon starts month (1); Full moon near mid-month (1).	1+1+1+1+1
36	Add the readings: 480 cm (1); Average = 480 / 4 = 120 cm (1); Repeating reduces random error (1); Readings vary a little (1); Average is more reliable (1).	1+1+1+1+1
	OR: Groups give protection, e.g. deer (1); Group hunting, e.g. wolves (1); Some animals live alone, e.g. tiger (1); Need large territory (1); Depends on food and safety (1).	1+1+1+1+1
37	Heat destroys vitamin C (1); Boiling vegetables loses water-soluble vitamins (1); Steaming retains nutrients (1); Cooking makes some foods digestible (1); Do not overcook (1).	1+1+1+1+1
	OR: Both attracted by the bar (1); Magnet may be repelled (1); Iron is not repelled (1); Test confirms magnet (1); Iron may become magnetised (1).	1+1+1+1+1
38	Passenger at rest wrt train (1); Trees seem to move backward (1); Sun seems to move across the sky (1); Earth's rotation causes it (1); Depends on frame (1).	1+1+1+1+1
	OR: Dissolve salt in water (1); Fold filter paper (1); Pour mixture (1); Sand stays on paper (1); Salt solution passes (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 3.5 km	Divide by 1000.
2	**(b)** salt and sugar	Both dissolve.
3	**(a)** below 35 °C	So the reading is accurate.
4	**(a)** boiling	Vaporisation.
5	**(d)** residue	It stays on paper.
6	**(a)** seed is living	Growth.
7	**(c)** oxygen	Combustion needs oxygen.
8	**(c)** Venus: hottest planet	Right.
9	**(a)** placing the ruler's zero mark at one end and reading straight on	Correct use of the scale.
10	**(c)** to reduce heat loss	Insulation.
11	**(a)** spinach	It provides iron and vitamins.
12	**(c)** aluminium	Aluminium is not attracted.
13	**(b)** circular motion	Path is circular.
14	**(a)** wax melts quickly along the rod when one end is heated	Heat travels along it.
15	**(b)** bulb is sensitive to body heat	Hand heat changes the reading.
16	**(b)** ice – solid	Ice is solid.
17	**(b)** crystallises out	Solubility falls.
18	**(c)** water	Aquatic eggs.
19	**(a)** aluminium cans	Metal recycled.
20	**(a)** rotate once on its axis	Rotation period.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	It collects light from far objects (1); It makes faraway objects look bigger and brighter (1).	1+1
22	Scales protect the body (1); Scales help the body to move smoothly through water (1).	1+1
23	Difficulty seeing in dim light (1); It may lead to night blindness (1).	1+1
24	They repel (1); Like poles repel (1).	1+1
25	Rotation: spinning about own axis, like Earth in a day (1); Revolution: going around another body, like Earth around the Sun (1).	1+1
26	No privacy (1); Frosted translucent glass is better (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	T = 42/21 = 2 s (1); 60 s / 2 s (1); 30 oscillations (1).	1+1+1
28	Rough mat cleans shoes (1); Smooth top is easy to clean (1); Window lets in light (1).	1+1+1
29	Shiny walls reflect heat back into the flask (1); Less heat escapes (1); Keeps drink hot or cold (1).	1+1+1
30	Only water evaporates (1); Salts remain (1); Condensed water is fresh (1).	1+1+1
31	Large surface area (1); Sun heats quickly (1); Faster evaporation (1).	1+1+1
	OR: Plants make food in leaves (1); Animals cannot (1); They depend on plants (1).	1+1+1
32	Do not decay (1); Block drains (1); Harm animals (1).	1+1+1
	OR: It orbits Earth (1); It was not made by humans (1); Artificial ones are man-made (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Hot water at 70 °C, cold water at 20 °C (1); Heat flows from hot to cold (1); Final in between (1); No more flow once equal (1); Same temperature (1).	1+1+1+1+1
34	Filter through cloth (1); Boil for 10 minutes (1); Cool (1); Store in a clean covered vessel (1); Kills germs (1).	1+1+1+1+1
35	Winnowing and sieving (1); Straining (1); Evaporation (1); Churning (1); Basic ideas same (1).	1+1+1+1+1
36	Plants: stomata (1); Fish: gills (1); Humans: lungs (1); All take in oxygen (1); Release carbon dioxide (1).	1+1+1+1+1
	OR: Population and vehicles rise (1); Fossil fuels deplete (1); Use solar (1); Use efficient devices (1); Save energy (1).	1+1+1+1+1
37	Water in liquid form (1); Oxygen (1); Right temperature (1); Protection by atmosphere (1); Distance from the Sun (1).	1+1+1+1+1
	OR: Form from observation: plants in shade look pale (1); Hypothesis: plants need sunlight (1); Test with plants in light and dark (1); Pale plants in dark support it (1); Repeat results, then accept (1).	1+1+1+1+1
38	Mango: net venation (1); Mango: tap root (1); Grass: parallel venation (1); Grass: fibrous roots (1); Reason: different groups of plants (1).	1+1+1+1+1
	OR: Starch: iodine on potato turns blue-black (1); Fat: butter leaves an oily patch on paper (1); Iodine on butter stays brown (1); Potato leaves no oily patch (1); Record in a table (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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